

Electronics & Circuits Bench

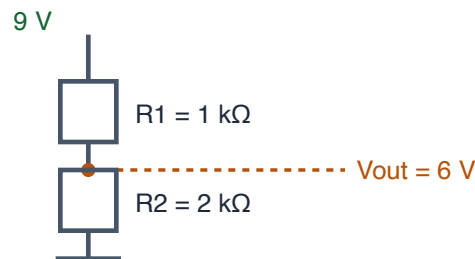
The full written lesson — Ohm's law, the bench, soldering, and debugging a dead circuit.

1. Ohm's law, and the one it hides

$V = I \times R$ is the whole of it, but the useful skill is reading which of the three a circuit is *setting* and which it is *letting happen*.

- A battery sets **V**. The resistance decides **I**.
- A resistor doesn't "limit current" by magic — it drops voltage, and the current follows.
- Power matters as much as the trio: $P = V \times I$. A ¼ W resistor passing 30 mA at 9 V is dissipating 0.27 W and will cook.

The trap: people memorise $V=IR$ and still wire an LED straight to 9 V. The LED is not a resistor — it holds roughly a fixed forward voltage (~2 V) and its current runs away. Without a series resistor, *I* is limited only by whatever the supply can deliver, which is how an LED dies in under a second. A test-writer's favourite distractor is "the LED has high resistance, so it is fine."



Worked example, drawn to match the numbers below.

Worked: 9 V across 1 kΩ + 2 kΩ in series. Total $R = 3 \text{ k}\Omega$, so $I = 9/3000 = 3 \text{ mA}$. The drop across $R2 = 0.003 \times 2000 = 6 \text{ V}$. That is the divider output — and note it only holds while nothing meaningful draws current from that midpoint.

2. The bench: multimeter and breadboard

Two instruments do most of the work, and both have a failure mode that looks like a broken circuit.

- **Voltage** is measured *across* a component — probes in parallel, circuit powered.
- **Current** is measured *through* it — the meter goes in series, which means breaking the circuit and putting the meter in the gap.
- **Continuity** beeps below a few ohms. Use it with the power **off**.

The trap: leaving the leads in the 10A / current jack and then measuring voltage. In current mode the meter is nearly a short circuit — across a supply it blows its fuse at best. Nearly every "dead multimeter" is a blown current fuse from exactly this.

Breadboards: the outer rails run the long way, the inner columns run in fives across a central gully. The gully exists so a DIP chip straddles it without shorting its own pins together.

Second trap: some breadboards *split* the power rails halfway along with no visible marking. Half your board is dead and the wiring looks perfect. Check the rails with continuity before believing anything.

3. Soldering: flux is the lesson

Solder does not stick to oxide, and every metal surface on your bench is oxidised. **Flux** is the chemical that strips that oxide at soldering temperature so molten solder *wets* the pad — flows out and bonds — instead of balling up and sitting on it.

- **Rosin-cored solder** carries flux in its core. That is usually enough for fresh boards.
- **Extra flux** (pen or paste) is for reworking, drag-soldering, and anything oxidised.
- **No-clean** leaves a residue that is safe to leave. **Water-soluble** flux is aggressive and **must** be washed off — left on, it corrodes the joint it just made.
- **Tin the tip**: a wetted tip transfers heat; a dry oxidised tip mostly does not, so you hold the iron there longer and lift pads.

The trap: "more heat" is the usual reaction to solder that will not flow, and it is usually the wrong one. Flux burns off in seconds — a tip parked on a joint has already destroyed the flux that was going to make it work, so more time makes it worse. Add flux, not seconds. Most beginner "bad solder" is a flux problem blamed on the iron.

Reading a joint: a good joint is concave and shiny — solder wetted both pad and lead and curved inward. A cold joint is dull and convex, a ball resting on the pad instead of bonded to it. A bridge spans two pads with too much solder. An insufficient joint leaves the pad edge still bare. Same pad and lead in all four cases — only the solder quantity and heat changes.

Temperature: ~330–370 °C for leaded, hotter for lead-free. **Ventilation**: the fumes are vaporised flux, not lead — still, do not breathe them. **Hygiene**: with leaded solder, wash your hands before eating. Every time.

4. Debugging a circuit that does nothing

A dead circuit is rarely a mystery; it is nearly always one of a short list, and the list is worth working in order rather than by intuition.

- **Power first.** Measure the rail *at the component*, not at the supply. A supply reading 5 V proves nothing about what reached pin 8.
- **Ground second.** Continuity from the chip's ground pin back to supply ground. A floating ground produces symptoms that look like anything at all.
- **Then orientation.** Electrolytic caps, diodes, LEDs and ICs are all directional. A backwards electrolytic can vent.
- **Then the joints.** Reflow anything dull. A cold joint is an intermittent connection, which is why the fault "comes and goes when I press on it".

The trap: changing two things at once. The circuit starts working and you have learned nothing — including whether one of your changes made something else worse. One change, one measurement, every time.

Worked: LED circuit does nothing. Rail at the LED anode = 9 V. Continuity cathode → ground = beeps. So power and ground are fine and the LED is the suspect: it is in backwards. Three measurements, no guessing, no parts replaced.